GENERAL SPECIFICATION G7.15

Fire Sprinkler System

CONTENTS

G7.15 FIRE SPRINKLER SYSTEM

G7.15.1 Scope

(a) This Section of the Specification contains requirements necessary to design, provide, and install, complete, fire water storage tanks, pumps, pressure maintenance (jockey) pumps, drivers, controllers, accessories, piping, and fittings.

(b) The Fire Water Storage Tanks and associated fire pumping system will form the reliable water supply source for the proposed water-based firefighting system and fire protection system. These assets are specified to serve as water supply source for fire mains.

(c) The Fire Water Storage Tanks will receive Domestic water supply feed from the PUB water mains with approved PUB bulk metering, piping, fittings and valves chamber facilities.

(d) An indicative location of the Fire Water Storage Tanks is shown on the general arrangement proposed for the site on the Drawings.

G7.15.2 Standards

(a) Work included in the Section shall conform to the following, except as specified otherwise within this Section:

Standard	Subject
NFPA 13	Sprinkler Systems
NFPA 20	Centrifugal Fire Pumps
NFPA 22	Standard for Water Tanks for Private Fire Protection

(b) Code of Practice for Fire Precautions in Buildings;

(c) Code on Accessibility in the Built Environment;

(d) Singapore Productivity and Standards Board (PSB):

Standard	Subject
CP 5	Code of Practice for Electrical Installations
CP48	Code of Practice for Water Services
CP52	Code of Practice for Automatic Fire Sprinkler System
SS508	Specification for Graphical Symbols – Safety Colours and Safety Signs;
SS532,	Code of Practice for The Storage of Flammable Liquids
SS575	ode of Practice for Fire Hydrant, Rising Mains and Hose Reel Systems

(e) FMRC Data Sheet 3-7N/13-4N, Fire Pumps;

(f) IRI Information Sheet IM.14.2.1, Centrifugal Fire Pumps; and

(g) Ministry of Manpower- Code of Practice for Workplace Safety and Health (WSH).

G7.15.3 Definitions

(a) Reference to a device or a part of the equipment applies to the number of devices or parts required to complete the installation.

(b) Churn Pressure refers to pump discharge pressure at zero flow.

G7.15.4 Design Criteria

(a) Performance

(i) Fire Pump: adjust setting for start at a pressure within 0.7bar of churn pressure;

(ii) Relief Valve: valve relief pressure 0.3bar greater than churn pressure;

(iii) Stopping Accuracy: plus or minus maximum 5mm under any loading condition; and

(iv) Jockey Pumps - Rated Capacity: size to exceed the flow from inspector's test valve without starting fire water pump.

(b) Equipment shall be UL listed and FM approved for fire protection services.

(c) Electrical driven sprinkler pump, jockey pump, and hose-reel pump shall be PSB approved and listed.

(d) Contractor shall file related documents, obtain and pay for permits for the portion of the fire sprinkler systems covered by the work, and arrange and pay for inspections and tests of the fire sprinkler system required by codes and standards and authorities having jurisdiction.

G7.15.5 Regulatory Requirements

(a) The fire system design shall comply with the most stringent applicable provisions of the Codes, ordinances, and laws applicable within Singapore, including revisions and changes in effect on date of these Specifications.

(b) The Contractor shall submit Hydraulic Calculations which demonstrate compliance with requirements of specified key references and codes, such as NFPA 22 and AWWA D100 or equal to comply with AHJ (Authorities Having Jurisdiction).

G7.15.6 Submittals

(a) The Contractor’s design proposal shall be designed by a Professional Engineer registered in Singapore who has proven experience in the design of works of this nature, and who is expressly engaged by the Contractor for the purpose of designing the fire water storage tank and pump system. The design plans shall be properly drawn to scale and prepared in a format suitable for submission to the Building and Construction Authority, Singapore and other relevant authorities.

(b) Submit datasheets from coating manufacturer as part of submittal.

(c) Provide the following submittals for the fire pump system:

(i) Detailed installation shop drawings of the complete pump system. Drawings shall show information required by NFPA 20 for working plans and include:

1) Pumps and drivers, jockey pumps;

2) Installation dimensions and cut lengths for piping;

3) Valves, fittings, and accessories; and

4) Sensing lines and controllers.

(ii) Technical data sheet and pump curves for impeller efficiency, NPSH available, absorbed brake horsepower, rpm, exhaust stack and diesel tank size.

(iii) Product data, including details on the pump, driver, controller, piping, and valving.

(iv) Product certificates signed by manufacturer.

(v) Bill of materials showing make, model, part number, and options.

(vi) Controller wiring interconnection and schematic, motor horsepower, and power supply voltage.

(vii) Full hydraulic calculation for most unfavourable and most favourable areas.

(viii) Certified fire pump performance curve.

(ix) Pump House Drawings: size, construction materials and drainage design. Operation and maintenance manuals, final drawings, and final calculations.

(x) Operation and maintenance data.

G7.15.7 Products

(a) General

(i) Manufacturers shall be listed in the latest published edition of Factory Mutual Research Corporation's Fire Protection Approval Guide.

(b) Fire Pumps

(i) Pump sets shall be of the centrifugal constant speed, single-stage, end suction vertical split case, direct-coupled to the electric motor, and be completed with mating flanges, air release valve, drain connections and base plates.

(ii) The pumps shall be selected for their particular applications with regard to water temperature, ambient air temperature and conditions, suction head, static head, lift, working pressure, and test pressure. Furthermore, the pump sets shall comply with all governing by-laws, regulations, order and requirements of all Local Authorities in respect of the various types of installations. Evidence of listing or approval by a recognised institution of the sprinkler pumps shall be provided.

(iii) Pumps casing shall be of close grained cast iron or cast steel suitable for the operating pressure and designed to provide smooth flow with the gradual changes in velocity. Impellers shall be of bronze and designed to give non-over loading characteristics over large range of head variations.

(iv) Each impeller shall be statically and dynamically balanced. The pump shafts shall be of machine ground stainless steel mounted in renewable bronze sleeve bearings. Pump glands shall be of the mechanical seal type but stuffing boxes with bronze glands will be considered as alternatives to the base offer as specified

(v) Performance

1) Each pump set shall be capable of circulating not less than the nominal rate specified in the Drawings against the system resistance.

2) The Contractor is required to carry out necessary calculations relating to the pressure drop of equipment selected by them, and of the piping of the system 80 that the pump set is capable of providing the necessary output at a designed head.

3) The pump shall be selected to operate close to the most efficient point on the pump curve.

4) Pump performance curves with the predicted operating point shown shall be submitted within eight weeks of the award of Contract. Nominal duties of the pump are given in the Schedule attached herein and shall be used as an approximate guide only.

5) The sprinkler pumps are to be factory tested to meet the specified performance and the requirements of the specified Code of Practice.

6) The pump sets are to be tested complete with the offered motor coupled to the pump. Documentary evidence of this testing shall be submitted for approval before the pump can be accepted. The motor couplings shall be of the flexible pin and bush type.

(vi) Speed

1) The motors shall be of drip proof, totally enclosed fan cooled (TEFC), squirrel cage type with a synchronous a 400V, 3 phase 50 Hz A.C supply. The insulation of the motor shall be of Class `F' to BS EN 60085 or equivalent. The torque/speed characteristic shall be selected to match the pump operation.

2) The Contractor shall produce manufacturer's selection charts and data to verify that direct coupling of the pump to the motor will meet the duly specified using standard impellers to the approval of the Consulting Engineer; otherwise, the Sub-Contractor shall provide at his own cost an indirect drive arrangement to the approval of the Consulting Engineer.

3) In general, the pump design, construction and fittings shall be compatible with the requirements for maximum reliability and minimum maintenance.

(vii) Pumps - Type:

Casing material	Cast Iron
Bearing housing material	Cast Iron
Casing bolts material	Steel
Bearing housing Gasket material	Rubber
Bearing housing bolts material	Steel
Bearing cap bolts material	Steel
Impeller material	Bronze
Impeller key material	Steel
Shaft material	Steel
Shaft sleeve material	Bronze
Shaft Seal Material	Graphited Acrylic Yarn
Motor	Suitable for Direct On Line Operation

1) The sprinkler pump sets shall comply with all requirements of the SAA Code.

2) Each pump set shall be capable of circulating not less than the nominal rate specified in the Drawings against the head of the system. The performance characteristics of the sprinkler pumps shall meet the requirements of the SAA Code.

3) The sprinkler pumps shall be capable of working at a pressure not less than 150% of the working pressure encountered in the system. Calculations of the system pressure and equipment manufacturer's undertaking shall be submitted for verification before the order of equipment.

4) The pumps must be capable of providing independently the necessary pressure and flows for the respective Hazard Class. The pumps should have performance characteristics such that the pressure falls progressively with the rate of demand. Pumps shall be selected to operate on the 'flat-point' on the flow characteristic curve. The pumps shall be equipped with automatic starting and to be automatically started based on pressure reduction in the system. Means should also be provided for automatic starting of the pumps by manually reproducing the pressure reduction. The pump must run continuously once started until being stopped manually and shall be manually reset after it has been activated. However, the control circuit shall provide automatic means of stopping the sprinkler jockey pump 30 seconds after the sprinkler system has attained its designed pressure.

5) The pump starter shall be of approved construction with the following minimum features:

- Incoming power supply isolator;
- Incoming supply `AVAILABLE' indication;
- Main isolator `ON' indication;
- Duty pump selector;
- Auto manual starting selector;
- Pump manual "ON OFF" push buttons;
- Pump "RUNNING" indication;
- Provision for sprinkler pump status (running) monitoring at the Main Alarm Panel (MAP) in accordance with the requirements of CP 10 and CP 52; and
- Provision for sprinkler pumps power failure indication and audible alarm at the Main Alarm Panel in accordance with the requirements of CP 10 & CP52.

6) Details of the pump starter panel complete with the control circuit and wiring connection drawing shall be submitted for the S.O.'s approval prior to construction. All pressure switches used shall be of approved type suitable for Fire Protection Installation Service.

7) For each pump installation, the following fittings are required as a minimum:

- Gate valve at suction;
- Globe valve at discharge;
- Check valve at discharge;
- Stainless steel flexible connection at suction & discharge;
- Eccentric tapered reducer at suction;
- Concentric tapered reducer at discharge;
- Strainer at suction;
- Listed pressure reducing valve or discharge branch-off to tank & listed pressure relief valve;
- Listed direct reading flow meter at branch-off to tank; and
- 100mm / 150mm dia. pressure gauge at suction & discharge.

8) All sprinkler pumps, including the jockey pump shall be provided with standby emergency power supply.

(viii) Jockey Pump Sets

1) All bye-laws, regulations and requirements of the local Fire Brigade Authorities and the Building Control Division, as well as SAA Code where connected to the sprinkler installations.

2) A jockey pump shall be provided in each set of sprinkler system. Unless otherwise specified, the jockey pump shall be of vertical in line type. The jockey pump shall be started automatically when the pressure drops in the system, due to normal leakage. It shall be automatically stopped when the system is reinstated. Notwithstanding what is stated in the drawings, the capacity and the head required shall be selected based on the actual system requirements.

3) Jockey pumps shall be capable of delivering a pump head of approximately 0.7 bar higher than the pressure in the trunk mains when the main pumps are churning. Pumps shall be carefully selected so as to operate on the steep flow characteristic curve and such that when delivering 180l/min., the pump head shall be around the pressure of the trunk main when the pump is churning.

4) The starter panel of the jockey pump shall generally follow the requirements of the sprinkler pump starter panel and shall be incorporated with the latter in one board

(ix) Installation Control Valves

1) The sprinkler installation shall be provided with installation control valve(s) as shown in the drawings. Each installation shall comprise:

- Main stop valve;
- Alarm valve c/w retard chamber;
- Water motor alarm & gong;
- Pressure gauges; and
- Drain valves and testing apparatus.

2) All to be supplied and installed conforming to the relevant Code of Practice and authorities' requirements.

(x) Drivers

1) Diesel Engine: water-cooled heat exchanger, and electric starter; and

2) Electric Motors: electric motor, totally enclosed. Comply with G5, OEM Electric Motors.

(xi) Accessories

1) Signal Contacts: Form C remote contacts, Fire Alarm and Smoke Detection Systems.

(xii) Alarms

1) Diesel Driver:

- Overheated engine;
- Low and high pump room temperature;
- Low fuel;
- Suction tank low level;

- Sequential timing device;
- Protected against overcurrent of 150 percent;
- Battery charger supply failure;
- Failure of engine to start automatically; and
- Diesel start circuit disconnected.

2) Electric Driver:

- Low and high pump room temperature;
- Sequential timing device; and
- Protected against overcurrent of 150 percent.

(xiii) Fuel Tank for Diesel Fire Pump: Comply with CP 52.

(xiv) Fuel Piping: Comply with G4, Carbon Steel Piping Systems.

(xv) Vibration isolation.

(xvi) Inlet and Exhaust System

1) Intake Air Filter: dry type;

2) Spark Arrestor/Silencer: residential noise-rated, heavy industrial type with flexible exhaust connection;

3) Sound Power Level: 45 dB(A) at 30m;

4) Silencer: sized for the engine with ANSI flanges and rain cap; and

5) Stainless steel (SS304) exhaust pipe of size not less than the engine exhaust outlet shall be installed from the engine exhaust connection to high level outside the pump room. The exhaust pipe shall be provided with high temperature insulation.

(xvii) Batteries shall comply with General Specification G5 – Electrical Works.

(xviii) Plain-End Pipe Fittings and Couplings: prohibited, including setscrew couplings.

(xix) Bushings and Grooved Reducing Couplings: prohibited.

(xx) Uniflanges: prohibited.

(c) Main Stop Valve

(i) Main stop valves shall be of the outside screw and yoke "right handed" type and shall be clearly marked showing direction which the hand wheel is to be turned to close the valve. Indication shall be provided to show whether the valve is in `open' or `shut' position. The valves shall be secured open by means of padlocked chains or straps to be provided in this Contract. Listed tamper-proof switches shall also be provided to monitor the valves status at the MAP.

(ii) An approved type notice with directional sign shall be provided on the outside of an external wall, as near to the various main stop valve locations as possible bearing the following words in raised letters or other approved type of letters:

SPRINKLER STOP VALVES INSIDE

(iii) The words "SPRINKLER STOP VALVES" shall be in letters of at least 35mm high and the word "INSIDE" in letters of at least 25mm high.

(iv) The words shall be painted white on red background.

(v) For each control valve, approved type labelling indicating Hazard Group, floor served, performance data, etc. shall be provided to the authority's requirements.

(d) Alarm Valve c/w Retard Chamber

(i) The alarm valve must be listed and complies with CP 52: Latest Edition. It shall be installed immediately above the main stop valve. The alarm valve shall have all necessary connections to pressure gauges, water motor alarm, combined drain and test valve, etc.

(e) Water Motor Alarm and Gong

(i) Water motor alarm shall be listed and complies to CP 52: Latest Edition. It shall comprise a fire alarm gong with a water turbine. The water motor alarm shall be suitably mounted on the wall above the alarm valve and all pipe connection shall not be less than 20mm diameter. The pipework work shall be galvanised and shall drain through a stainless steel fitting having an orifice not larger than 3mm diameter. Drain lines shall be connected to a turn dish, which shall be drained to nearest floor waste trap.

(f) Pressure Gauges

(i) Pressure gauges shall be fitted at the following position:

- 1 No. immediately above each alarm valve; and
- 1 No. immediately below each main stop valve

(ii) All tapping for pressure gauge connections shall be fitted with approved stop cocks and syphon tubes and means must be provided to allow each pressure gauge to be readily removed without interruption of water supply.

(iii) The pressure gauge shall be of not less than 100mm diameter and shall conform to BS EN 837. The scale of reading shall have a range of 150% of the known maximum pressure and with divisions not exceeding 0.2 bar for a maximum scale value of 10 bar, not exceeding 0.5 bar for a maximum scale value of 16 bar and not exceeding 1 bar for maximum scale value in excess of 16 bar.

(iv) All pressure gauges shall be of the heavy duty and weather proof type construction.

(g) Testing Apparatus

(i) Permanent testing apparatus shall be provided to enable proving test of water supplies to be carried out on completion.

(ii) The testing apparatus shall consist of Testing stop valve, together with listed shunt gap meter to indicate actual flow in l/min. Test pressure gauges shall be similarly be provided.

(iii) Size of the shunt gap meter shall be selected to match the expected flow to be tested in the system.

(h) Locking Strap

(i) Locking strap c/w lock of master key system shall be provided for all valves at:

- Suction/discharge of pump installation;
- Drain and main stop valve at control valves; and
- Valves at pressure test set of sprinkler & hose reel system.

(i) Spare Parts

(i) Provide one complete set of spare mechanical seals for each pump type.

(ii) Provide one complete set of spare bearings for each pump type

(j) Sprinkler Head

(i) Sprinklers shall be FM approved and UL listed.

(ii) Contractor shall supply and install all sprinkler heads as shown on the Schematic Drawings. The spacing and location of sprinklers shall meet with the specified Code of Practice requirements and should any part of the system shown in the Schematic Drawings deviated from the requirements, the S.O. should be immediately notified.

(iii) Contractor shall comply with SS CP52, with regards to maximum and minimum spacing between sprinklers on range pipes and between adjacent rows of sprinklers, maximum distance of sprinklers from walls or partitions, maximum distance of sprinkler heads below ceiling or roofs, minimum horizontal distance of sprinklers from beams, etc.

(iv) Easily accessible flushing facilities shall be provided for each zone to discharge to the nearest drainage outlet.

(v) Sprinklers shall be new and of approved makes, types and temperature ratings. Sprinklers shall be conventional pattern designed to produce a spherical type of discharge with a proportion of water being thrown upwards to the ceiling; sidewall pattern type shall be used where appropriate. Sprinklers shall be designed with universal type deflector enabling the sprinkler to be erected in either the upright or pendent position.

(vi) Heavy duty approved type sprinkler guards shall be provided by the Contractor in situations where sprinkler is liable to accidental or mechanical damage.

(vii) The sizes of sprinkler orifices for various hazard class and the temperature ratings for various areas shall meet the requirements of the specified Code of Practice and of the FSSD or other relevant Authority.

(viii) Protective guards of approved type shall be fitted to each sprinkler head in the area where sprinklers are located less than 2.0 metres above floor level and where they are liable to accidental and mechanical damaged. Sprinkler heads installed in lift shafts and pits shall be fitted with protective guards. Where applicable or required, these sprinkler heads shall be of the side-throw type.

(ix) For concealed sprinkler heads and concealed pipework installation (that is, installed inside false ceiling space), sprinklers shall be of standard bronze finished and installed preferably in the upright position.

(x) For exposed sprinkler heads (that is, installed at the suspended ceiling) and concealed pipework installations (that is, installed within false ceiling space), sprinklers shall be installed pendent complete with ceiling plate flush to the suspended ceiling. Sprinklers shall be of satin chromed finish to S.O.'s requirements. The Contractor should allow for fixed deflectors type and not retracted deflectors. Ceiling plates shall be of stainless steel 304 construction and shall be either painted to S.O.'s requirements.

(xi) For exposed sprinkler heads and exposed pipework installation, where no suspended ceiling is used, sprinklers shall be installed preferably pendent position and shall be stainless steel 304 finished to S.O.'s requirement.

(k) Orifice Plates

(i) Orifice plates shall be complete with matching flanges properly spaced, drilled and gasketed and complete with flange bolts. Tapping points shall be provided both upstream and downstream of the orifice plates and shall be made in strict accordance with BS 1042 and associated amendments.

(ii) Orifice plates shall be fitted as necessary in order to assist in hydraulically balancing the system or to meet pump characteristic curve. The orifice plates shall be of approved type and shall be installed in accordance with SAA Code.

(l) Breeching Inlet

(i) The breeching inlets shall be approved by the local authorities and listed under the Product Listing Scheme.

(ii) The fire brigade breeching inlet shall be:

- 2 inlets for 100mm dia. riser; and
- 4 inlets for 150mm dia. riser.

(iii) Inlets shall be of 65mm dia. instantaneous male coupling type with non-return valves and chained cups.

(iv) Inlets shall have a 25mm dia. drain valve with 25mm dia. drainpipe connected to nearest floor trap.

(v) Earthing is to be provided for the breeching inlet at the inlet connection.

(vi) Stainless steel 304 breeching inlet cabinets enclosure is to be provided for the breeching inlet and to be earthed.

(m) Flow Switch

(i) An approved type of flow switch shall be provided and connected to each alarm valve for initiation of an audible and visible alarm signal at the Fire Alarm Panels on activation of the sprinkler system. The wiring works to the Main Alarm Panel shall be included in this Contract. Spare voltage free contacts (N/O + N/C) shall be made available on the flow switch to be used by others for remote indication.

(ii) The flow switch shall have single pole double throw (SPDT) mechanism which makes or breaks the electric circuit when water flows. All components of the flow switch that come in contact with the water shall be made of polyethylene or polypropylene.

(iii) Contacts shall be suitable for the working voltage and current of the circuits controlled and shall be made of silver or approved alloy.

(iv) Adequate space shall be allowed above the pipework for the installation of flow switches.

(v) Flow switches of the self-resetting type shall be installed in accordance with the manufacturer's recommendations.

(vi) Flows switches shall incorporate retards or time-delay devices to avoid false alarms due to surges.

(vii) The electrical contact block shall be completely sealed from the water in the pipeline.

(viii) If additional flow switches are required in the distribution pipelines as indicated in the Drawings, they shall be provided.

(ix) Where flow switches are installed within ceiling spaces, access panels shall be provided to facilitate maintenance of the flow switch. Flow switches for sprinkler pipes installed at high ceiling spaces shall be installed at low levels for ease of maintenance. The location of each flow switch shall be clearly identified on the access panel. The flow switch interfacing module shall be provided with enclosure of IP rating 55.

(x) Test valves shall be installed immediately downstream of each flow switch to enable local testing. Short drain pipe shall be provided after each test valve and properly terminated with cap. The test valves shall be accessible from the floor level.

(xi) Flow switches shall be of vane type LPC or UL/FM listed.

(n) Fire Water Tanks

(i) The Contractor shall supply, deliver and install water storage tanks as shown in the drawings.

(ii) Where stainless steel water tanks are shown in the drawings they shall be stainless steel 304 sectional type complete with cover, stays, cleats, stainless steel bolts, nuts and washers and jointing compound complying with SS 22, compartmented and with capacities not less than that indicated in the drawings.

(iii) Each tank shall be installed above floor slab and located in the positions shown in the Drawing and shall be mounted on dwarf walls in concrete or R.S.J. beam.

(iv) All tank plates shall be 4.8mm thick. Tank shall be erected in accordance with manufacturer's instructions and all joints made with approved jointing compound. The tank covers shall not be less than 3mm thick supported on angle bearers cleared to the tank plate flanges. A 600mm manhole with hinged lid shall be provided to each compartment of all tanks.

(v) Aluminium ladders of approved type shall be provided inside and outside of water tanks to facilitate access into the compartment.

(vi) Each tank compartment shall be provided with valve inlet, outlet and drain connections, non-modulating type of pilot float valve, vortex inhibitor, overflow connection, water level indicators, manholes, etc. as shown in the drawings

(vii) All drain pipe ends, vents and overflow pipe ends shall be provided with suitable anti-vermin screens.

(viii) All overflow warning pipes shall be extended to out-side of the pump room, or acceptable location to achieve the intended purpose.

(ix) Transparent tube level indicators shall be provided for each of the tank compartment. The tube shall be of heavy duty construction with metric calibration marks in red for the entire tank height. Isolation valves and drain cocks shall be provided at the bottom of the pipe.

(o) Floatless Switch High and Low Level Alarm System

(i) The make-up or transfer pumps shall be automatically controlled by stainless steel electrode and floatless switch relays.

(ii) The electrodes shall be installed at the sprinkler water storage tank(s). The relay control panel shall be installed at the pump room or as shown in the drawings.

(iii) The high and low level control and alarm system shall have the following features.

1) The electrodes at the storage tanks shall start and stop the duty pump at pre-determined levels i.e. `start pump' & `stop pump' levels at the storage tanks and `stop pump' level at the make-up tanks.

2) The low level alarm shall be activated when the water level falls to the pre-determined `danger' low level.

3) The high level alarm shall be activated when the water level rises to the pre-determined `danger' high level.

4) A klaxon or siren shall be provided above the door of each pump room for audible alarm indication. Remote alarm indication shall also be provided to locations as shown in the drawings.

(iv) The relay control panel shall not be limited to the following:

1) High and low level alarm indication;

2) Audible alarm muting push button;

3) Alarm reset push button;

4) Control system "ON OFF" switch; and

5) 1 NO and 1 NC spare voltage free contacts for both high and low alarm relays to enable future connections for remote alarm indication.

The electrodes shall be of the water contact low voltage type of stainless steel construction.

G7.15.8 Cleaning

(a) Prior to shipping, perform final cleaning of internal and external surfaces after manufacturing and testing operations have been completed.

G7.15.9 Execution

(a) Installation

(i) Install each equipment item in accordance with manufacturer's directions, referenced codes, and specifications;

(ii) Alignment Procedures shall comply with NFPA 20, appendix item A-3-5 and other relevant standard under clause G7.15.2;

(iii) Install items so they may be easily removed for maintenance and repair;

(iv) Install items so that access for maintenance is safe and readily available; and

(v) Fuel Tank Installation shall comply with NFPA 20, Figure A-8-4.6. Fuel Tank Low-Level Sensor alarm shall be set at minimum level allowed by NFPA 20.

(b) Testing

(i) Acceptance Tests: Comply with NFPA 13 and 20, including NFPA 20, appendix items A-11-2.6.3 and A-11-2.6.3(f), Test Procedure, and requirements of the authority having jurisdiction.

(ii) Fire water tanks to be tested in place by sealing connections, filling with water, and allowing water to stand in tank for minimum of 24 hours. Any leaks including visible wet patches, or defects which may cause leakage, shall be rectified and re-tested until the tank is completely watertight. If a leak is detected while the tank is being filled with water, the Contractor may repair the defect responsible for the leak before continuing filling the tank to the top water level. Filling and emptying of the tank shall be carried out by the Contractor. A copy of the hydrostatic test results shall be submitted to the S.O.

END OF SECTION